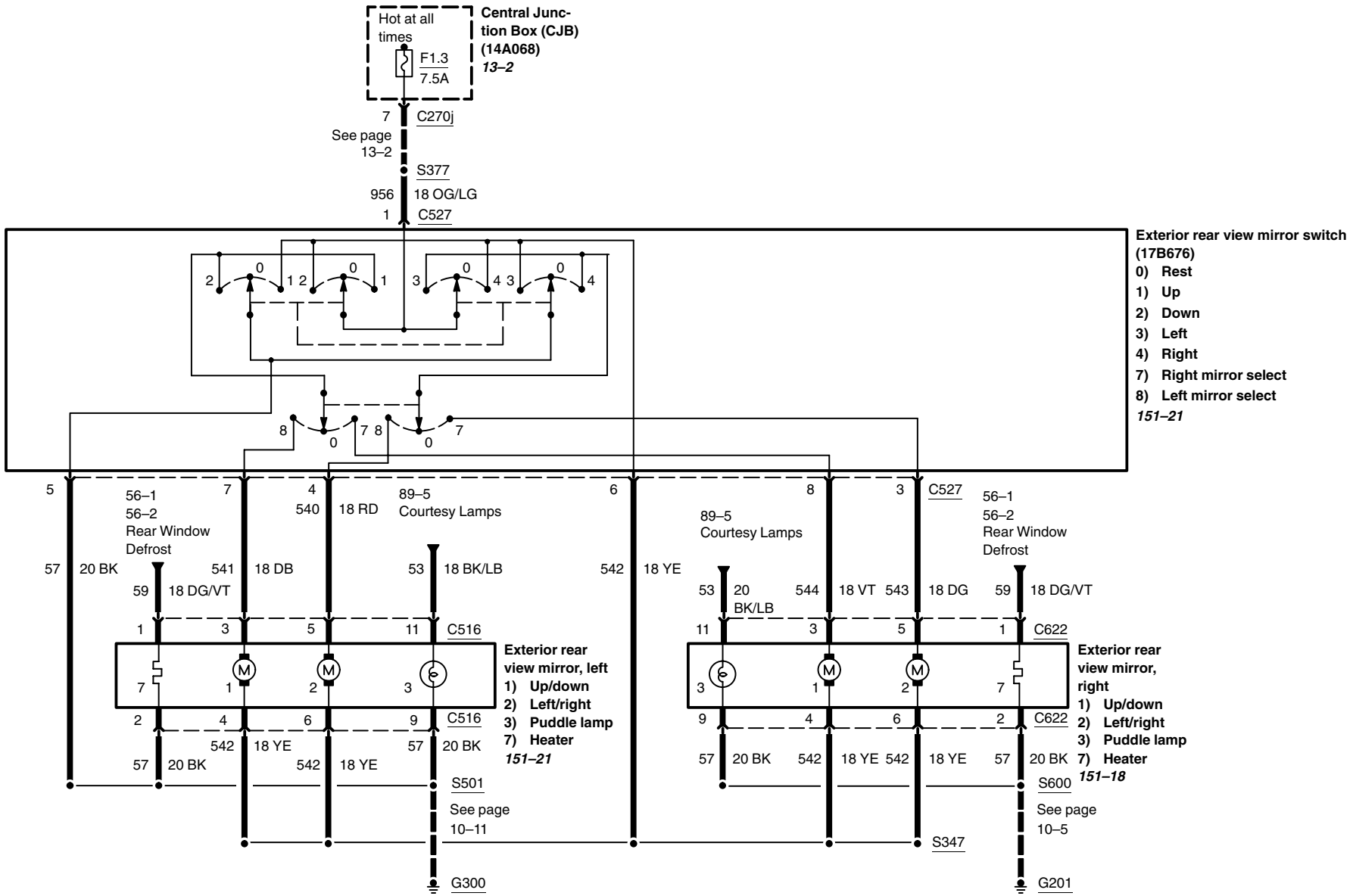## without memory